

Azhar Abdool

Software & Computer Engineer | Applied AI / Machine Learning / Data

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SUMMARY

Software Engineer at NCT Forestry and UCT Computer Science and Computer Engineering graduate, building enterprise applications, database workflows and integrations. Developing an applied AI and ML focus through PyTorch classification, reinforcement learning, scikit-learn data-mining pipelines and Palantir Foundry work. Combines model-development foundations with practical backend, data and secure application engineering. Currently completing part-time Honours study in Security & Network Engineering.

TECHNICAL SKILLS

- **AI / ML:** Python, PyTorch, scikit-learn, supervised classification, tabular Q-learning, K-Means, PCA, model evaluation, preprocessing and feature selection.
- **Data / Enterprise AI:** pandas, NumPy, SQL, Oracle, PL/SQL, Palantir Foundry, Ontology, AIP Logic, Gemini integration, data transformation and analytical workflows.
- **Software / Integration:** Java, JavaScript, C, C++, C#, React Native, Oracle APEX, REST APIs, Firebase, Git, testing and debugging.
- **Systems / Security:** STM32, ADC/PWM, DMA, concurrency, TCP/UDP, RBAC, input validation and application security; Azure security course training.

EXPERIENCE

Software Engineer | NCT Forestry

October 2025 - Present

- Build and maintain enterprise PIMS modules using PL/SQL, Oracle APEX and React Native.
- Develop SQL reporting queries, backend integrations and operational workflow logic supporting enterprise data applications.
- Implement database procedures, role-based access control and input validation for secure application workflows.
- Develop GIS-enabled applications integrating Google Maps, GeoJSON and polygon validation with PL/SQL backend logic.
- Collaborate through Git workflows, debugging, testing and deployments; resolve geospatial rendering and integration issues.

SAP Graduate Programme | OpenSource Intelligent Solutions

October 2024 - March 2025

- Completed SAP Associate training and earned the Business Process Integration with SAP S/4HANA certification.
- Participated in enterprise systems workshops and SAP technical sessions, with ABAP on SAP BTP and ERP process learning records.

SELECTED AI / ML PROJECTS

Student Performance Data Mining | Python, scikit-learn, pandas | 2026

- Developed supervised classification pipelines comparing Logistic Regression and Decision Tree models, with train-only imputation, encoding, feature selection and a stratified 80/20 evaluation split.
- Removed identifier and direct outcome-leakage columns; reproduced 94.75% Logistic Regression test accuracy on the supplied dataset during local verification.
- Applied scaled K-Means clustering and PCA to academic performance/engagement data, alongside EDA and SQL reporting. Results are coursework evaluations, not deployed or externally validated predictions.

MNIST Neural Network Classifier | Python, PyTorch | 2024

- Built a 784-128-64-10 feedforward network with grayscale normalisation, Adam minibatch training, test evaluation and local image prediction.
- Recorded 95.71% MNIST test accuracy in the original training log; added reproducible setup controls during portfolio cleanup.

Four Rooms Reinforcement Learning | Python, NumPy | 2024

- Implemented tabular Q-learning scenarios for package collection using state-action values, shaped rewards and deterministic/stochastic transitions.
- Produced recorded agent-path visualisations and documented state-bookkeeping and convergence limitations; added a bounded execution wrapper during cleanup.

Foundry AI Infographic Dashboard | Palantir Foundry, Ontology, AIP Logic, Gemini | 2026

- Designed Foundry data transformations and Ontology-backed analytical workflows for enterprise KPI exploration.
- Implemented AIP Logic natural-language interactions and Gemini-generated infographic summaries, with dashboard, query and generation-history interfaces.

EDUCATION

BSc Honours Security & Network Engineering | Eduvos

Current, part-time online study.

BSc Computer Science & Computer Engineering | University of Cape Town

Graduated 2025. Software engineering, algorithms, databases, networks, embedded systems and AI/ML. Earlier Mechatronics study before transferring.

CERTIFICATIONS AND TRAINING

- **Ontology University - Junior Forward Deployed Engineer:** Certificate of Achievement, issued June 2026.
- **SAP Certified Associate - Business Process Integration with SAP S/4HANA:** earned January 2025; credential expired January 2026.
- **Microsoft AZ-500, SC-200 and SC-5002:** Azure security, security operations and Defender for Cloud course/training records.
- **SAP ABAP on SAP BTP / Basic ERP Processes:** learning records and training, 2024-2025.